

CURRICULUM VITAE

ROLAND KWITT

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Birth Date	March 19, 1982	
Citizenship	Austrian	
Academic Details	<i>h</i> -index: 39, Cites: 6134 (Source: Google Scholar, 04/2026)	
Webpage	https://rkwitt.github.io/	
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CURRENT EMPLOYMENT

- 2022 - now **University of Salzburg**
Full Professor (for Machine Learning, § 99 (4) UG)
Department of Artificial Intelligence and Human Interfaces (AIHI)
Jakob-Haringer Str. 2, A-5020 Salzburg, Austria
- 2020 - 2021 **University of Salzburg**
Full Professor (for Machine Learning, § 99 (4) UG)
Department of Computer Science
Jakob-Haringer Str. 2, A-5020 Salzburg, Austria
- 2017 - 2020 **University of Salzburg**
Associate Professor
Department of Computer Science
Jakob-Haringer Str. 2, A-5020 Salzburg, Austria
- 2013 - 2017 **University of Salzburg**
Assistant Professor
Department of Computer Science
Jakob-Haringer Str. 2, A-5020 Salzburg, Austria

PREVIOUS EMPLOYERS

- 2011 - 2013 **Kitware Inc.**
R & D Engineer, Computer Vision / Medical Imaging Group
101 E Weaver St., NC 27510, USA
Supervisor(s): Stephen Aylward, Brad Davis

EDUCATION

- 2010 - 2011 *PostDoc*, CS department, University of Salzburg (ADVISOR(S): Andreas Uhl, Wolfgang Pree)
- 2007 - 2010 *Dr. techn. (equiv. to PhD)*, CS department, University of Salzburg (ADVISOR: Andreas Uhl)
- 2005 - 2007 *Dipl.-Ing. (equiv. to MSc)*, CS department, University of Salzburg (ADVISOR: Ulrich Hofman)
- 2001 - 2005 *Dipl.-Ing. (FH) (equiv. MSc)*, Telecommunications Engineering, University of Applied Sciences Salzburg (ADVISOR: Ulrich Hofman)

AWARDS

- 2014 **CVPR '14 Outstanding Reviewer**
- 2012 Short-listed for the "Heinz-Zemanek" price 2012 (notification e-mail upon request)
- 2012 **MICCAI '12 Young Investigator Award**, awarded at MICCAI '12 (Nice, France)
- 2007 **Best Paper Award**, International Conference on Computer Recognition Systems (CORES '07)
- 2005 **Special Appreciation Award**, Austrian Ministry of Science and Research

MAIN RESEARCH AREAS

Machine learning, Computer vision, Medical image analysis

RANKING IN SELECTION PROCESSES FOR *Full Professorships*

- Ranked **1st** for § 98 UG professorship *Data Science* at the University of Innsbruck (“Ruf” received 12/2019)
- Ranked **1st** for § 99 UG professorship *Machine Learning / Data Science* at the University of Klagenfurt (“Ruf” received 06/2020)

THIRD-PARTY FUNDING (GRANTED)

- 2025-2028 *doc.funds.connect REVELATION: Artificial Intelligence driven Biomedical Imaging Innovation* (granted)
(joint project with the FH Salzburg as project lead; 6 PhD positions)
Funding source: FWF & Land Salzburg
Project volume € 1.152.216,50.-
- 2023-2025 *servEB: Globales Daten- und Patientenmanagement bei seltenen, genetischen Hautkrankheiten* (ongoing)
(joint project with the Paracelsus Medical University and the Salzburger Landeskliniken (Lead))
Funding source: Amt der Salzburger Landesregierung
Project volume € 478.000,00.- (University of Salzburg share: € 100.000,00.-)
- 2023 - 2027 *Intelligent Data Analytics (IDA) Lab III* (ongoing)
(together with W. Trutschnig (Head), C. Borgelt, and A. Bathke, Department of Artificial Intelligence and Human Interfaces, University of Salzburg)
Funding source: Amt der Salzburger Landesregierung (within the WISS¹ initiative)
Project volume € 1,600.000,00.-
- 2019 - 2023 *Intelligent Data Analytics (IDA) Lab I & II*
(together with W. Trutschnig (Head), Department of Mathematics, University of Salzburg and C. Borgelt, Department of Mathematics/Computer Science, University of Salzburg)
Funding source: Amt der Salzburger Landesregierung (within the WISS¹ initiative)
Project volume € 2,300.000,00.-
- 2020 - 2022 *Free-of-Bias, Robust and Intelligent Data Analytics (FRIDA)* (completed)
(together with W. Trutschnig, Department of Mathematics, University of Salzburg)
Funding source: Porsche Holding GmbH
Industry partner: Porsche Informatik GmbH
Project volume: € 250.000,00.-
- 2019 - now *Deep Homological Learning* (completed)
Funding source: FWF (Project Nr. P 31799)
Project volume: € 238.512,75
- 2019 - 2022 *Kundenfokussierte Zukunftstrends (KFZ)* (completed)
(together with W. Trutschnig, Department of Mathematics, University of Salzburg)
Funding source: Land Salzburg (within the WISS 2025 initiative)
Industry partner: Porsche Informatik GmbH
Project volume: € 478.610,38.-
- 2018 - 2021 *“Kleinprojekte” Critical data & Feature selection* (completed)
(together with W. Trutschnig, Department of Mathematics, University of Salzburg)
Funding source: Porsche Informatik GmbH
Industry partner: Porsche Informatik GmbH
Project volume: € 40,000.-
- 2018 *Synonym Analysis for Improving Search Queries* (completed)
(together with N. Augsten, Department of Computer Science, University of Salzburg)

¹Wissenschafts- und Innovationsstrategie Salzburg

Industry partner: FindoLogic GmbH
Funding source: FFG (Innovationsscheck 5,000)
Project volume: € 5,000.-

2018 *Data Analytics in Industrial Environments* (completed)
(together with W. Trutschnig, Department of Mathematics, University of Salzburg)
Industry partner: Siemens Austria
Project volume: € 20,000.-

STUDENT SUPERVISION

PRIMARY PHD/POSTDOC ADVISOR

- Dominik Geng (PhD student, since 2025)
- Martin Uray (PhD student, since 2024, co-supervised with Stefan Huber)
- Günther Eder (PostDoc completed, now in industry)
- Sebastian Zeng (PhD completed, now PostDoc at Linnaeus University, Sweden)
- Florian Graf (PhD completed, now PostDoc in my group)
- Christoph D. Hofer (PhD & PostDoc completed, now in industry)

SECONDARY PHD ADVISOR

- Mann Willi (PhD completed, now in industry)
- Wimmer Georg (PhD completed, now PostDoc at University of Salzburg)
- Kauba Christof (PhD completed, now PostDoc at University of Salzburg)
- Debiasi Luca (PhD completed, now in industry)
- Ribeiro Eduardo (PhD completed)
- González Tejeda Yansel (PhD ongoing)
- Höller Yvonne (PhD ongoing)
- Schraml Rudolf (PhD ongoing)
- Kirchgasser Simon Ignaz (PhD ongoing)

MSc ADVISOR

- Johanna Wald
- Schmitzberger Nina Marie
- Michael Kastner
- Söllinger Dominik
- Grafendorfer Philipp
- Tobias Hilgart
- Philip Brandauer
- Peer Raphael
- Marlene Holzleitner
- Sabine Hasenleithner

PUBLICATIONS (IN REVERSE-CHRONOLOGICAL ORDER)

For conference articles, the (most recent) ICORE ranking is provided as **A***, **A**, **B**,... (if available). Similarly, for journal articles, the journal impact factor (**IF**) is listed as well (if available).

JOURNAL ARTICLE (PEER-REVIEWED)

- [Dyc+21] L.E. van Dyck, R. Kwitt, S.J. Denzler, and W.R. Gruber. "Comparing object recognition in humans and deep convolutional neural networks - An eye tracking study". In: *Front. Neurosci* 15 (2021). DOI: [10.3389/fnins.2021.750639](https://doi.org/10.3389/fnins.2021.750639). **IF: 3.2**
- [C H+20] C. Hofer, R. Kwitt, Y. Höller, E. Trinka, and A. Uhl. "An empirical assessment of appearance descriptors applied to MRI for automated diagnosis of TLE and MCI". In: *Comput. Biol. Med.* 117 (2020). DOI: [10.1016/j.combiomed.2019.103592](https://doi.org/10.1016/j.combiomed.2019.103592). **IF: 6.3**
- [CRM19] C. Hofer, R. Kwitt, and M. Niethammer. "Learning Representations of Persistence Barcodes". In: *J. Mach. Learn. Res.* 20.126 (2019), pp. 1–45. **IF: 4.1**
- [D R+19] D. R. Chittajallu, M. McCormick, S. Gerber, T.J. Czernuszcwicz, R. Gessner, M.S. Willis, M. Niethammer, R. Kwitt, and S.R. Aylward. "Image-Based Methods for Phase Estimation, Gating, and Temporal Superresolution of Cardiac Ultrasound". In: *IEEE Trans. Biomed. Eng.* 66.1 (2019), pp. 72–79. DOI: [10.1109/TBME.2018.2823279](https://doi.org/10.1109/TBME.2018.2823279). **IF: 4.5**
- [N S+19] N. Stanley, T. Bonacci, R. Kwitt, M. Niethammer, and P.J. Mucha. "Stochastic Block Models with multiple continuous attributes". In: *Appl. Netw. Sci.* 4.54 (2019). DOI: [10.1007/s41109-019-0170-z](https://doi.org/10.1007/s41109-019-0170-z). **IF: 1.5**
- [Z D+19] Z. Ding, G. Fleishman, X. Yang, P. Thomson, R. Kwitt, and M. Niethammer. "Fast predictive simple geodesic regression". In: *Med. Image Anal.* 56 (2019), pp. 193–209. DOI: [10.1016/j.media.2019.06.003](https://doi.org/10.1016/j.media.2019.06.003). **IF: 11.8**
- [N S+18] N. Stanley, R. Kwitt, M. Niethammer, and P.J. Mucha. "Compressing Networks with Super Nodes". In: *Nature Sci. Rep.* 8.10892 (2018). DOI: [DOI:10.1038/s41598-018-29174-3](https://doi.org/10.1038/s41598-018-29174-3).
- [X H+18] X. Han, R. Kwitt, S.R. Aylward, S. Bakas, B. Menze, A. Asturias, P. Vespa, J. van Horn, and M. Niethammer. "Brain extraction from normal and pathological images: A joint PCA/Image-Reconstruction approach". In: *NeuroImage* 176.8 (2018), pp. 431–445. DOI: [10.1016/j.neuroimage.2018.04.073](https://doi.org/10.1016/j.neuroimage.2018.04.073). **IF: 5.7**
- [Yan+17] X. Yang, R. Kwitt, M. Styner, and M. Niethammer. "Quicksilver: Fast Predictive Image Registration - a Deep Learning Approach". In: *NeuroImage* 158 (2017), pp. 378–396. DOI: [10.1016/j.neuroimage.2017.07.008](https://doi.org/10.1016/j.neuroimage.2017.07.008). **IF: 5.7**
- [Hon+16a] Y. Hong, R. Kwitt, N. Singh, N. Vasconcelos, and M. Niethammer. "Parametric Regression on the Grassmannian". In: *IEEE Trans. Pattern Anal. Mach. Intell.* 38.11 (2016). DOI: [10.1109/TPAMI.2016.2516533](https://doi.org/10.1109/TPAMI.2016.2516533). **IF: 18.6**
- [Liu+15a] X. Liu, M. Niehammer, R. Kwitt, N. Singh, M. McCormick, and S. Aylward. "Low-Rank Atlas Image Analyses in the Presence of Pathologies". In: *IEEE Trans. Med. Imaging* 34.12 (2015), pp. 2583–2591. DOI: [10.1109/TMI.2015.2448556](https://doi.org/10.1109/TMI.2015.2448556). **IF: 10.0**
- [Hon+14c] Y. Hong, B. Davis, J. S. Marron, R. Kwitt, N. Singh, J. S. Kimbell, E. Pitkina, R. Superfine, S.D. Davis, C. J. Zdanski, and M. Niethammer. "Statistical atlas construction via weighted functional boxplots". In: *Med. Image Anal.* 18.4 (2014), pp. 684–698. DOI: [10.1016/j.media.2014.03.001](https://doi.org/10.1016/j.media.2014.03.001). **IF: 11.8**
- [Kwi+13b] R. Kwitt, N. Vasconcelos, S. Razzaque, and S. Aylward. "Localizing Target Structures in Ultrasound Video - A Phantom Study". In: *Med. Image Anal.* 17.7 (2013), pp. 712–722. DOI: [10.1016/j.media.2013.05.003](https://doi.org/10.1016/j.media.2013.05.003). **IF: 11.8**
- [Kwi+12b] R. Kwitt, N. Vasconcelos, N. Rasiwasia, A. Uhl, B. Davis, M. Häfner, and F. Wrba. "Endoscopic Image Analysis in Semantic Space". In: *Med. Image Anal.* 16.7 (2012), pp. 1415–1422. DOI: [10.1016/j.media.2012.04.010](https://doi.org/10.1016/j.media.2012.04.010). **IF: 11.8**
- [KMU11a] R. Kwitt, P. Meerwald, and A. Uhl. "Efficient Texture Image Retrieval Using Copulas in a Bayesian Framework". In: *IEEE Trans. Image Process.* 20.7 (2011), pp. 2063–2077. DOI: [10.1109/TIP.2011.2108663](https://doi.org/10.1109/TIP.2011.2108663). **IF: 13.7**

- [KMU11b] R. Kwitt, P. Meerwald, and A. Uhl. “Lightweight Detection of Additive Watermarking in the DWT-Domain”. In: *IEEE Trans. Image Process.* 20.2 (2011), pp. 474–484. DOI: [10.1109/TIP.2010.2064327](https://doi.org/10.1109/TIP.2010.2064327). IF: 13.7
- [KU10a] R. Kwitt and A. Uhl. “Lightweight Probabilistic Texture Retrieval”. In: *IEEE Trans. Image Process.* 19.1 (2010), pp. 241–253. DOI: [10.1109/TIP.2009.2032313](https://doi.org/10.1109/TIP.2009.2032313). IF: 13.7
- [Haf+09a] M. Häfner, R. Kwitt, A. Uhl, A. Gangl, F. Wrba, and A. Vécsei. “Feature-Extraction from Multi-Directional Multi-Resolution Image Transformations for the Classification of Zoom-Endoscopy Images”. In: *Pattern Anal. Appl.* 12.4 (2009), pp. 407–413. DOI: [10.1007/s10044-008-0136-8](https://doi.org/10.1007/s10044-008-0136-8). IF: 2.0
- [Haf+08a] M. Häfner, R. Kwitt, A. Uhl, A. Gangl, F. Wrba, and A. Vécsei. “Computer-assisted Pit-Pattern Classification in Different Wavelet Domains for Supporting Dignity Assessment of Colonic Polyps”. In: *Pattern Recognit.* 42.6 (2008), pp. 1180–1191. DOI: [10.1016/j.patcog.2008.07.012](https://doi.org/10.1016/j.patcog.2008.07.012). IF: 7.6

CONFERENCE ARTICLES (PEER-REVIEWED)

- [F G+25] F. Graf, P. Pellizzoni, M. Uray, S. Huber, and R. Kwitt. “The Flood Complex: Large-Scale Persistent Homology on Millions of Points”. In: *NeurIPS*. 2025. URL: <https://openreview.net/forum?id=tT3EQdFhtX>. A*
- [Gre+25] H. Greer, L. Tian, F.-X. Vialard, R. Kwitt, R.S.J. Estepar, and M. Niethammer. “CARL: A Framework for Equivariant Image Registration”. In: *CVPR*. 2025. A*
- [Dem+24] B. Demir, L. Tian, H. Greer, R. Kwitt, F.-X. Vialard, R.S.J. Estepar, S. Bouix, R. Rushmore, E. Ebrahim, and M. Niethammer. “MultiGradICON: A Foundation Model for Multimodal Medical Image Registration”. In: *WBIR*. 2024. DOI: [10.1007/978-3-031-73480-9_1](https://doi.org/10.1007/978-3-031-73480-9_1).
- [L T+24] L. Tian, H. Greer, R. Kwitt, F.-X. Vialard, R.S.J. Estepar, S. Bouix, R. Rushmore, and M. Niethammer. “uniGradICON: A Foundation Model for Medical Image Registration”. In: *MICCAI*. 2024. A
- [Pap+24] T. Papamarkou, T. Birdal, M. Bronstein, G. Carlsson, J. Curry, Y. Gao, M. Hajj, R. Kwitt, P. Lio, P. Di Lorenzo, V. Maroulas, N. Miolane, F. Nasrin, K.N. Ramamurthy, B. Rieck, S. Scardapane, M.T. Schaub, P. Velickovic, B. Wang, Y. Wang, G.-W. Wei, and G. Zamzmi. “Position Paper: Challenges and Opportunities in Topological Deep Learning”. In: *ICML*. 2024. A*
- [Zen+24] S. Zeng, F. Graf, M. Uray, S. Huber, and R. Kwitt. “Neural Persistence Dynamics”. In: *NeurIPS*. 2024. A*
- [Gre+23] H. Greer, L. Tian, F.-X. Vialard, R. Kwitt, S. Bouix, Estepar R.S.J., R.J. Rushmore, and Marc Niethammer. “Inverse Consistency by Construction for Multistep Deep Registration”. In: *MICCAI*. 2023. DOI: [10.1007/978-3-031-43999-5_65](https://doi.org/10.1007/978-3-031-43999-5_65). A
- [Tia+23] L. Tian, H. Greer, F.-X. Vialard, Estepar R.S.J., R.J. Rushmore, N. Makris, S. Bouix, and Marc Niethammer. “GradICON: Approximate Diffeomorphisms via Gradient Inverse Consistency”. In: *CVPR*. 2023. URL: <https://tinyurl.com/48e99fxj>. A*
- [ZGK23] Sebastian Zeng, Florian Graf, and Roland Kwitt. “Latent SDEs on Homogeneous Spaces”. In: *NeurIPS*. 2023. URL: <https://tinyurl.com/56s99s8s>. A*
- [Gra+22] F. Graf, S. Zeng, B. Rieck, M. Niethammer, and R. Kwitt. “On Measuring Excess Capacity in Neural Networks”. In: *NeurIPS*. 2022. URL: <https://tinyurl.com/3yue6mzn>. A*
- [F G+21] F. Graf, C. Hofer, M. Niethammer, and R. Kwitt. “Dissecting Supervised Contrastive Learning”. In: *ICML*. 2021. URL: <https://tinyurl.com/2u4wtwm4>. A*
- [H G+21] H. Greer, R. Kwitt, F.-X. Vialard, and M. Niethammer. “ICON: Learning Regular Maps Through Inverse Consistency”. In: *ICCV*. 2021. DOI: [10.1109/ICCV48922.2021.00338](https://doi.org/10.1109/ICCV48922.2021.00338). A*
- [ZGK21] S. Zeng, F. Graf, and C. Hofer R. Kwitt. “Topological Attention for Time Series Forecasting”. In: *NeurIPS*. 2021. URL: <https://tinyurl.com/2eg5yoyh>. A*
- [C H+20a] C. Hofer, F. Graf, B. Rieck, M. Niethammer, and R. Kwitt. “Graph Filtration Learning”. In: *ICML*. 2020. URL: <https://tinyurl.com/2mmrwqnm>. A*

- [C H+20b] C. Hofer, F. Graf, M. Niethammer, and R. Kwitt. "Topologically Densified Distributions". In: *ICML*. 2020. URL: <https://tinyurl.com/2gjfelru>. A*
- [F-X+20] F.-X. Vialard, R. Kwitt, S. Wei, and M. Niethammer. "A Shooting Formulation of Deep Learning". In: *NeurIPS*. 2020. URL: <https://tinyurl.com/2zqaz3pa>. A*
- [C H+19] C. Hofer, R. Kwitt, M. Dixit, and M. Niethammer. "Connectivity-Optimized Representation Learning via Persistent Homology". In: *ICML*. 2019. URL: <https://tinyurl.com/2erfwbte>. A*
- [MF19] M. Niethammer and R. Kwitt F.-X. Vialard. "Metric Learning for Image Registration". In: *CVPR*. 2019. DOI: 10.1109/CVPR.2019.00866. A*
- [Liu+18a] B. Liu, M. Dixit, R. Kwitt, and N. Vasconcelos. "Feature Space Transfer for Data Augmentation". In: *CVPR*. 2018. DOI: 10.1109/CVPR.2018.00947. A*
- [Gre+18] H. Greer, S. Gerber, M. Niethammer, R. Kwitt, M. McCormick, D. Chittajallu, N. Siekierski, M. Oetgen, K. Cleary, and S. Aylward. "Scoliosis Screening and Monitoring Using Self Contained Ultrasound and Neural Networks". In: *ISBI*. 2018. DOI: 10.1109/ISBI.2018.8363857.
- [Dix+17] M. Dixit, R. Kwitt, M. Niethammer, and N. Vasconcelos. "AGA: Attribute-Guided Augmentation". In: *CVPR*. 2017. DOI: 10.1109/CVPR.2017.355. A*
- [Han+17] X. Han, X. Yang, R. Kwitt, and M. Niethammer. "Efficient Registration of Pathological Images: A joint PCA/Image-Reconstruction Approach". In: *ISBI*. 2017. DOI: 10.1109/ISBI.2017.7950456.
- [Hof+17a] C. Hofer, R. Kwitt, Y. Höller, E. Trinka, M. Niethammer, and A. Uhl. "Constructing Shape Spaces from a Topological Perspective". In: *IPMI*. 2017. DOI: 10.1007/978-3-319-59050-9_9.
- [Hof+17b] C. Hofer, R. Kwitt, Y. Höller, E. Trinka, and A. Uhl. "Simple Domain Adaptation for Cross-Dataset Analyses of Brain MRI Data". In: *ISBI*. 2017. DOI: 10.1109/ISBI.2017.7950556.
- [Hof+17c] C. Hofer, R. Kwitt, M. Niethammer, and A. Uhl. "Deep Learning with Topological Signatures". In: *NIPS*. 2017. URL: <https://bit.ly/2UHfsCf>. A*
- [Hon+17] Y. Hong, X. Yang, R. Kwitt, M. Styner, and M. Niethammer. "Regression Uncertainty on the Grassmannian". In: *AISTATS*. 2017. URL: <https://bit.ly/2HL9nB0>. A
- [Yan+17] X. Yang, R. Kwitt, M. Styner, and M. Niethammer. "Fast Predictive Multimodal Image Registration". In: *ISBI*. 2017. DOI: 10.1109/ISBI.2017.7950652.
- [Gad+16a] M. Gadermayr, S. Hegenbart, R. Kwitt, and A. Uhl. "Narrow Band Imaging Versus White-Light: What is best for Computer-Assisted Diagnosis of Celiac Disease?" In: *ISBI*. 2016. DOI: 10.1109/ISBI.2016.7493282.
- [KHN16a] R. Kwitt, S. Hegenbart, and M. Niethammer. "One-Shot Learning of Scene Locations via Feature Trajectory Transfer". In: *CVPR*. 2016. DOI: 10.1109/CVPR.2016.16.
- [Ayl+16a] S. Aylward, M. McCormick, H.J. Kang, S. Razzaque, R. Kwitt, and M. Niethammer. "Ultrasound Spectroscopy". In: *ISBI*. 2016. DOI: 10.1109/ISBI.2016.7493437.
- [Yan+16] X. Yang, X. Han, E. Park, S. Aylward, R. Kwitt, and M. Niethammer. "Registration of Pathological Images". In: *Proceedings of the MICCAI Workshop on Simulation and Synthesis in Medical Imaging*. 2016. DOI: 10.1007/978-3-319-46630-9_10.
- [YKN16] X. Yang, R. Kwitt, and M. Niethammer. "Fast Predictive Image Registration". In: *Proceedings of the MICCAI Workshop on Deep Learning in Medical Image Analysis*. 2016. DOI: 10.1007/978-3-319-46976-8_6.
- [Kwi+15a] R. Kwitt, S. Huber, M. Niethammer, W. Lin, and U. Bauer. "Statistical Topological Data Analysis – A Kernel Perspective". In: *NIPS*. 2015. URL: <https://bit.ly/2ucS0B9>.
- [Rei+15a] R. Reininghaus, U. Bauer, S. Huber, and R. Kwitt. "A Stable Multi-scale Kernel for Topological Machine Learning". In: *CVPR*. 2015. DOI: 10.1109/CVPR.2015.7299106. A*
- [Hon+15a] Y. Hong, N. Singh, R. Kwitt, and M. Niethammer. "Group Testing for Longitudinal Data". In: *IPMI*. 2015. DOI: 10.1007/978-3-319-19992-4_11.
- [HKN15a] Y. Hong, R. Kwitt, and M. Niethammer. "Model Criticism for Regression on the Grassmannian". In: *MICCAI*. 2015. DOI: 10.1007/978-3-319-24574-4_87. A

- [Hon+14a] Y. Hong, N. Singh, R. Kwitt, and M. Niethammer. "Time-warped Geodesic Regression". In: *MICCAI*. 2014. DOI: [10.1007/978-3-319-10470-6_14](#). **A**
- [Kwi+14a] R. Kwitt, S. Razzaque, J. Lowell, and S. Aylward. "Variability sensitivity of dynamic texture based recognition in clinical CT data". In: *SPIE Medical Imaging*. 2014. DOI: [10.1117/12.2043271](#).
- [Liu+14a] X. Liu, M. Niethammer, R. Kwitt, M. McCormick, and S. Aylward. "Low-Rank to the Rescue: Atlas-based Analyses in the Presence of Pathologies". In: *MICCAI*. 2014. DOI: [10.1007/978-3-319-10443-0_13](#). **A**
- [Heg+14a] S. Hegenbart, R. Kwitt, N. Rasiwasia, A. Vécsei, and A. Uhl. "Do We need Annotation Experts? A Case Study in Celiac Disease Classification". In: *MICCAI*. 2014. DOI: [10.1007/978-3-319-10470-6_57](#). **A**
- [Hon+14b] Y. Hong, R. Kwitt, N. Singh, B. Davis, and M. Niethammer. "Geodesic Regression on the Grassmannian". In: *ECCV*. 2014. DOI: [10.1007/978-3-319-10605-2_41](#).
- [Hon+13a] Y. Hong, B. Davis, J.S. Marron, R. Kwitt, and M. Niethammer. "Weighted Functional Boxplot with Application to Statistical Atlas Construction". In: *MICCAI*. 2013. DOI: [10.1007/978-3-642-40760-4_73](#). **A**
- [Kwi+13a] R. Kwitt, D. Pace, M. Niethammer, and S. Aylward. "Studying Cerebral Vasculature Using Structure Proximity and Graph Kernels". In: *MICCAI*. 2013. DOI: [10.1007/978-3-642-40763-5_66](#). **A**
- [KVR12a] R. Kwitt, N. Vasconcelos, and N. Rasiwasia. "Scene Recognition on the Semantic Manifold". In: *ECCV*. 2012. DOI: [10.1007/978-3-642-33765-9_26](#).
- [Kwi+12a] R. Kwitt, N. Vasconcelos, S. Razzaque, and S. Aylward. "Recognition in Ultrasound Videos: Where Am I?" In: *MICCAI*. 2012. DOI: [10.1007/978-3-642-33454-2_11](#). **A**
- [Gsc+11a] M. Gschwandtner, R. Kwitt, W. Pree, and A. Uhl. "Infrared Camera Calibration for Dense Depth Map Construction". In: *IV*. 2011. DOI: [10.1109/IVS.2011.5940515](#).
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- [Kwi+11a] R. Kwitt, N. Rasiwasia, N. Vasconcelos, A. Uhl, M. Häfner, and F. Wrba. "Learning Pit Pattern Concepts for Gastroenterological Training". In: *MICCAI*. 2011. DOI: [10.1007/978-3-642-23626-6_35](#). **A**
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- [Haf+09b] M. Häfner, A. Gangl, R. Kwitt, A. Uhl, A. Vécsei, and F. Wrba. "Improving Pit-Pattern Classification of Endoscopy Images by a Combination of Experts". In: *MICCAI*. 2009. DOI: [10.1007/978-3-642-04268-3_31](#). **A**
- [Heg+09c] S. Hegenbart, R. Kwitt, M. Liedlgruber, A. Uhl, and A. Vécsei. "Impact of Duodenal Image Capturing Techniques and Duodenal Regions on the Performance of Automated Diagnosis of Celiac Disease". In: *ISPA*. 2009. DOI: [10.1109/ISPA.2009.5297637](#).
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- [KMU09c] R. Kwitt, P. Meerwald, and A. Uhl. "Efficient Detection of Additive Watermarking in the DWT-Domain". In: *EUSIPCO*. 2009. URL: <http://goo.gl/1S4c1A>.
- [KMU09b] R. Kwitt, P. Meerwald, and A. Uhl. "Blind DT-CWT Domain Additive Spread-Spectrum Watermark Detection". In: *DSP*. 2009. DOI: [10.1109/ICDSP.2009.5201255](#).
- [KMU09a] R. Kwitt, P. Meerwald, and A. Uhl. "Color-Image Watermarking using Multivariate Power-Exponential Distribution". In: *ICIP*. 2009. DOI: [10.1109/ICIP.2009.5413715](#). **B**

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- [KU08a] R. Kwitt and A. Uhl. “Image Similarity Measurement by Kullback-Leibler Divergences between Complex Wavelet Subband Statistics for Texture Retrieval”. In: *ICIP*. 2008. DOI: [10.1109/ICIP.2008.4711909](https://doi.org/10.1109/ICIP.2008.4711909). **B**
- [KU07a] R. Kwitt and A. Uhl. “Modeling the Marginal Distributions of Complex Wavelet Coefficient Magnitudes for the Classification of Zoom-Endoscopy Images”. In: *MMBIA*. 2007. DOI: [10.1109/ICCV.2007.4409170](https://doi.org/10.1109/ICCV.2007.4409170).

THESES

- [Kwitt10a] R. Kwitt. “Statistical Modeling in the Wavelet Domain and Applications”. PhD thesis. Department of Computer Science, University of Salzburg, Austria, 2010.

CONFERENCE TALKS & PRESENTATIONS

- 12/2025 *The Flood Complex: Large-Scale Persistent Homology on Millions of Points*, NeurIPS '25, San Diego, USA
- 12/2024 *Neural Persistence Dynamics*, NeurIPS '24, Vancouver, CA
- 12/2023 *Latent SDEs on Homogeneous Spaces*, NeurIPS '23, New Orleans, USA
- 12/2022 *On Measuring Excess Capacity in Neural Networks*, NeurIPS '22, New Orleans, USA
- 12/2021 *Topological Attention for Time Series Forecasting*, NeurIPS '21 (Virtual conference)
- 06/2021 *Dissecting Supervised Contrastive Learning*, ICML '21 (Virtual conference)
- 12/2020 *A Shooting Formulation of Deep Learning*, NeurIPS '20 (Virtual conference)
- 07/2020 *Topologically Densified Distributions*, ICML '20, Vienna, Austria (Virtual conference)
- 07/2020 *Graph Filtration Learning*, ICML '20, Vienna, Austria (Virtual conference)
- 09/2019 *Deep Homological Learning*, ÖMG Tagung, Dornbirn, Austria
- 06/2019 *Metric Learning for Image Registration*, CVPR '19, Long Beach, CA, USA
- 06/2016 *One-Shot Learning of Scene Locations via Feature Trajectory Transfer*, CVPR '16, Las Vegas, USA
- 12/2015 *Statistical Topological Data Analysis – A Kernel Perspective*, NIPS '15, Montreal, Canada
- 10/2015 *Model Criticism for Regression on the Grassmannian*, MICCAI '15, Munich, Germany
- 06/2015 *A Stable Multi-Scale Kernel for Topological Machine Learning*, CVPR '15, Boston, USA
- 09/2014 *Geodesic Regression on the Grassmannian*, ECCV '14, Zurich, Switzerland
- 09/2014 *Do we need Annotation Experts – A Case Study in Celiac Disease Classification*, MICCAI '14, Boston, USA
- 09/2014 *Low-Rank to the Rescue – Atlas-based Analyses in the Presence of Pathologies*, MICCAI '14, Boston, USA
- 10/2013 *Studying Cerebral Vasculature Using Structure Proximity and Graph Kernels*, MICCAI '13, Nagoya, Japan
- 10/2012 *Scene Recognition on the Semantic Manifold*, ECCV '12, Florence, Italy
- 10/2012 *Recognition in US Video: Where Am I?*, MICCAI '12, Nice, France
- 09/2011 *Learning Pit Pattern Concepts for Gastroenterological Training*, MICCAI '11, Toronto, Canada
- 08/2010 *Statistical Modeling in the Wavelet Domain and Applications*, PhD defense, Salzburg, Austria
- 11/2009 *A Joint Model of Complex Wavelet Coefficients for Texture Retrieval*, ICIP '09, Cairo, Egypt
- 11/2009 *Color-Image Watermarking using Multivariate Power-Exponential Distribution*, ICIP '09, Cairo, Egypt
- 09/2009 *Improving Pit Pattern Classification by a Combination of Experts*, MICCAI '09, London, UK
- 10/2008 *Image Similarity Measurement by Kullback-Leibler Divergences between Complex Wavelet Subband Statistics for Texture Retrieval*, ICIP '08, San Diego, CA, USA
- 04/2008 *Color Eigen-Subband Features for Endoscopy Image Classification*, ICASSP '08, Las Vegas, NV, USA
- 10/2007 *Modeling the Marginal Distributions of Complex Wavelet Coefficient Magnitudes for the Classification of Zoom-Endoscopy Images*, MMBIA '07, Rio de Janeiro, Brazil

INVITED TALKS

- 04/2022 *Topologically Densified Distributions*
ICLR Workshop on Geometrical and Topological Representation Learning 2022 (Virtual)
- 09/2021 *Topological Machine Learning*

Thematic Mini-Conference on Computational Topology and Machine Learning (Virtual)
Berlin Mathematics Research Center

- 09/2019 *Deep Homological Learning*
ÖMG Conference 2019, Dornbirn (in the *Computational Geometry and Topology* session)
- 01/2018 *Machine Learning with Topological Signatures*
Oberwolfach Workshop “Statistics for Data with Geometric Structure”, Oberwolfach, Germany
- 04/2016 *Low rank to the Rescue: Atlas-based Analyses in the Presence of Pathologies*
“Images and Networks of the Brain”, Hamburg, Germany
- 07/2015 *Topological Machine Learning*
ISNPS '15, Graz Austria
- 04/2014 *Grassmannian Geodesic Regression*
IST Austria, Austria
- 06/2013 *Localizing Target Structures In Ultrasound Videos*
Quantitative Medical Imaging (QMI), Arlington, VA, USA
- 12/2012 *Scene Recognition on the Semantic Manifold*
SVCL, UC San Diego, USA
- 10/2012 *Recognition in US Video: Where Am I?*
University of North Carolina, Chapel Hill (Computer Science), NC, USA

TEACHING

Except for a small selection of undergraduate courses, all lectures/labs are taught in English.

- SS 26 Seminar Multimedia Technologies (SE, graduate level), University of Salzburg
- SS 26 Statistical Learning Theory (VO+PS, graduate level), University of Salzburg
- WS 25/26 AI Lab / AI Werkstatt (UV+SE, undergraduate level), University of Salzburg
- WS 25/26 Machine Learning (VO, undergraduate level), University of Salzburg
- WS 25/26 Introduction to AI / AI Eingangswerkstatt (VO, undergraduate level), University of Salzburg
- WS 25/26 Computer Vision (VO+PS, graduate level), University of Salzburg
- SS 25 Statistical Learning Theory (VO+PS, graduate level), University of Salzburg
- SS 25 Seminar Multimedia Technologies (SE, graduate level), University of Salzburg
- WS 24/25 AI Lab / AI Werkstatt (UV+SE, undergraduate level), University of Salzburg
- WS 24/25 Machine Learning (VO, undergraduate level), University of Salzburg
- WS 24/25 Introduction to AI / AI Eingangswerkstatt (VO, undergraduate level), University of Salzburg
- WS 24/25 Interpreting and Presenting Statistical Analyses (SE, graduate level), University of Salzburg
- WS 24/25 Case Studies (SE, graduate level), University of Salzburg
- WS 24/25 Computer Vision (VO+PS, graduate level), University of Salzburg
- SS 24 Statistical Learning Theory (VO+PS, graduate level), University of Salzburg
- SS 24 Seminar Multimedia Technologies (SE, graduate level), University of Salzburg
- WS 23/24 Machine Learning (VO, undergraduate level), University of Salzburg
- WS 23/24 Introduction to AI / AI Eingangswerkstatt (VO, undergraduate level), University of Salzburg
- WS 23/24 Interpreting and Presenting Statistical Analyses (SE, graduate level), University of Salzburg
- WS 23/24 Case Studies (SE, graduate level), University of Salzburg
- WS 23/24 Seminar Multimedia Technologies (SE, graduate level), University of Salzburg
- WS 23/24 Computer Vision (VO+PS, graduate level), University of Salzburg
- SS 23 Statistical Learning Theory (VO+PS, graduate level), University of Salzburg
- SS 23 Databases 1 (VO+PS, undergraduate level), University of Salzburg
- SS 23 Medical Imaging (VO+PS, graduate level), University of Salzburg
- SS 23 Seminar Multimedia Technologies (SE, graduate level), University of Salzburg
- WS 22/23 Introduction to AI / AI Eingangswerkstatt (VO, undergraduate level), University of Salzburg
- WS 22/23 Interpreting and Presenting Statistical Analyses (SE, graduate level), University of Salzburg
- WS 22/23 Case Studies (SE, graduate level), University of Salzburg
- WS 22/23 Seminar Multimedia Technologies (SE, graduate level), University of Salzburg
- WS 22/23 Computer Vision (VO+PS, graduate level), University of Salzburg
- SS 22 Statistical Learning Theory (VO+PS, graduate level), University of Salzburg
- SS 22 Databases 1 (VO+PS, undergraduate level), University of Salzburg
- SS 22 Medical Imaging (VO+PS, graduate level), University of Salzburg
- SS 22 Seminar Multimedia Technologies (SE, graduate level), University of Salzburg

WS 21/22 Interpreting and Presenting Statistical Analyses (SE, graduate level), University of Salzburg

WS 21/22 Case Studies (SE, graduate level), University of Salzburg

WS 21/22 Introduction to Data Science (SE, graduate level), University of Salzburg

WS 21/22 Seminar Multimedia Technologies (SE, graduate level), University of Salzburg

WS 20/21 Computer Vision (VO+PS, graduate level), University of Salzburg

SS 21 Statistical Learning Theory (VO+PS, graduate level), University of Salzburg

SS 21 Databases 1 (VO+PS, undergraduate level), University of Salzburg

SS 21 Imaging Beyond Consumer Cameras (VO+PS, graduate level), University of Salzburg

SS 21 Seminar Multimedia Technologies (SE, graduate level), University of Salzburg

WS 20/21 Interpreting and Presenting Statistical Analyses (SE, graduate level), University of Salzburg

WS 20/21 Case Studies (SE, graduate level), University of Salzburg

WS 20/21 Introduction to Data Science (SE, graduate level), University of Salzburg

WS 20/21 Seminar Multimedia Technologies (SE, graduate level), University of Salzburg

WS 20/21 Computer Vision (VO+PS, graduate level), University of Salzburg

SS 20 Statistical Learning Theory (VO+PS, graduate level), University of Salzburg

SS 20 Databases 1 (VO+PS, undergraduate level), University of Salzburg

SS 20 Imaging Beyond Consumer Cameras (VO+PS, graduate level), University of Salzburg

SS 20 Seminar Multimedia Technologies (SE, graduate level), University of Salzburg

WS 19/20 Interpreting and Presenting Statistical Analyses (SE, graduate level), University of Salzburg

WS 19/20 Case Studies (SE, graduate level), University of Salzburg

WS 19/20 Introduction to Data Science (SE, graduate level), University of Salzburg

WS 19/20 Seminar Multimedia Technologies (SE, graduate level), University of Salzburg

WS 19/20 Computer Vision (VO+PS, graduate level), University of Salzburg

SS 19 Imaging Beyond Consumer Cameras (VO+PS, graduate level), University of Salzburg

SS 19 Seminar Multimedia Technologies (SE, graduate level), University of Salzburg

SS 19 Statistical Learning Theory (VO+PS, graduate level), University of Salzburg

SS 19 Databases 1 (PS, undergraduate level), University of Salzburg

WS 18/19 Seminar Multimedia Technologies (SE, graduate level), University of Salzburg

WS 18/19 Computer Vision (VO+PS, graduate level), University of Salzburg

WS 18/19 Introduction to Data Science (SE, graduate level), University of Salzburg

WS 18/19 Case Studies (SE, graduate level), University of Salzburg

WS 18/19 BSc Seminar (SE, undergraduate level), University of Salzburg

SS 18 Imaging Beyond Consumer Cameras (VO+PS, graduate level), University of Salzburg

SS 18 Seminar Multimedia Technologies (SE, graduate level), University of Salzburg

SS 18 Statistical Learning Theory (VO+PS, graduate level), University of Salzburg

SS 18 Databases 1 (PS, undergraduate level), University of Salzburg

SS 18 BSc Seminar (SE, undergraduate level), University of Salzburg

WS 17/18 Wissenschaftliche Arbeitstechniken (VP, undergraduate level)

WS 17/18 Image Processing and Computer Vision (VO+PS, undergraduate level), University of Salzburg

WS 17/18 Case Studies (SE, graduate level), University of Salzburg

WS 17/18 Computer Vision (VO+PS, graduate level), University of Salzburg

WS 17/18 Introduction to Data Science (SE, graduate level), University of Salzburg

SS 17 Seminar Multimedia Technologies (SE, graduate level), University of Salzburg

SS 17 Statistical Learning Theory (VO+PS, graduate level), University of Salzburg

SS 17 Computer Science for Everyone (VO, undergraduate level), University of Salzburg

SS 17 Databases 1 (VO+PS, undergraduate level), University of Salzburg

SS 17 BSc Seminar (SE, undergraduate level), University of Salzburg

WS 16/17 Introduction to Operating Systems (VO, undergraduate level), University of Salzburg

WS 16/17 Computer Vision (VO+PS, graduate level), University of Salzburg

WS 16/17 Introduction to Data Science (SE, graduate level), University of Salzburg

WS 16/17 Seminar Multimedia Technologies (SE, graduate level), University of Salzburg

SS 16 Imaging Beyond Consumer Cameras (VO+PS, graduate level), University of Salzburg

SS 16 Seminar Multimedia Technologies (SE, graduate level), University of Salzburg

SS 16 Databases 1 (PS, undergraduate level), University of Salzburg

WS 15/16 Seminar Multimedia Technologies (SE, graduate level), University of Salzburg

WS 15/16 Advanced Image Processing & Computer Vision (VO+PS, graduate level), University of Salzburg

SS 15 Seminar Multimedia Technologies (SE, graduate level), University of Salzburg

SS 15 Databases 1 (PS, undergraduate level), University of Salzburg

SS 15	Statistical Learning Theory (VO+PS, graduate level), University of Salzburg
WS 14/15	Seminar Multimedia Technologies (SE, graduate level), University of Salzburg
WS 14/15	Advanced Image Processing & Computer Vision (VO+PS, graduate level), University
SS 14	Databases 1 (PS, undergraduate level), University of Salzburg
SS 14	Imaging Beyond Consumer Cameras (VO+PS, graduate level), University of Salzburg
SS 14	Seminar Multimedia Technologies (SE, graduate level), University of Salzburg
WS 13/14	Advanced Image Processing & Computer Vision (VO+PS, graduate level), University of Salzburg
	Introduction to Object Oriented Programming (VO, undergraduate level), FH Salzburg
	Network Management (UE, undergraduate level), FH Salzburg

SERVICE TO THE UNIVERSITY OF SALZBURG

2022 - now	Head of the <i>Curricularkommission BA Artificial Intelligence</i>
2022 - now	Head of the division <i>Computer Vision and Machine Learning (CVML)</i>
2022 - now	Deputy head of the department of <i>Artificial Intelligence and Human Interfaces (AIHI)</i>
2016 - 2024	Member of the <i>Curricularkommission MA Data Science</i>
2019 - 2021	Member of the <i>Curricularkommission BA/MA Computer Science</i>
2019	Appointment committee member for the <i>Assistant Professor position in Database Systems</i>
2017 - 2018	Appointment committee member for the <i>§ 99 UG professorship for Data Science ("Stiftungsprofessur")</i>

PROFESSIONAL SERVICE

Area Chair (AC) for ICML '22, ICML '23, ICML '24, ICML '25, ICML '26, NeurIPS '21, NeurIPS '22, NeurIPS '23, NeurIPS '24, NeurIPS '25, NeurIPS '26, AAAI '25, ICLR '25, ICLR '26
 General Chair of the 39th OAGM/ AAPR Workshop 2015, Salzburg, Austria
 Organizer of the 3rd Biannual Austrian TDA Meeting 2022, Salzburg, Austria
 PC Chair of ACM IH & MMSEC 2014, Salzburg, Austria

JOURNAL REVIEWING

Reviewer for *Journal of Machine Learning Research (JMLR)*
 Reviewer for *IEEE Transactions on Medical Imaging (TMI)*
 Reviewer for *IEEE Transactions on Image Processing (TIP)*
 Reviewer for *IEEE Transactions on Signal Processing*
 Reviewer for *IEEE Signal Processing Letters*
 Reviewer for *Elsevier Medical Image Analysis (MedIA)*
 Reviewer for *Foundations of Computational Mathematics (FOCM)*
 Reviewer for *Journal of Applied and Computational Topology (APCT)*

CONFERENCE REVIEWING

Reviewer for *International Conference on Learning Representations (ICLR)*
 Reviewer for *International Conference on Machine Learning (ICML)*
 Reviewer for *Artificial Intelligence and Statistics (AISTATS)*
 Reviewer for *Neural Information Processing Systems (NeurIPS)*
 Reviewer for *IEEE International Conference on Image Processing (ICIP)*
 Reviewer for *Medical Image Computing and Computer Assisted Intervention (MICCAI)*
 Reviewer for *International Conference on Computer Vision (ICCV)*
 Reviewer for *Computer Vision and Pattern Recognition (CVPR)*
 Reviewer for *European Conference on Computer Vision (ECCV)*
 Reviewer for *British Machine Vision Conference (BMVC)*
 Reviewer for *International Conference on Pattern Recognition (ICPR)*

FUNDING AGENCIES

Reviewer for the *European Research Council (ERC)*

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More references available upon request.

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